Agentic AI Trading System

Automated Trading Workflow — Final Report

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Module: Agentic AI Workflow for Automated Trading (Finance Domain)

University: NIBM - Coventry University

GitHub: github.com/pasiuvi/agentic-trading-system

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This report presents a complete Agentic AI-based trading system covering data collection via Yahoo Finance API, AWS cloud integration (S3, RDS, EC2, Lambda), exploratory data analysis, Random Forest ML modelling, and a four-agent autonomous decision pipeline for five financial assets.

1. Executive Summary

This project designs and evaluates an Agentic AI-based automated trading workflow. The system autonomously collects financial market data, performs EDA, trains Random Forest ML models, retrieves live news sentiment via VADER NLP, and generates buy/sell/hold decisions through a four-agent sequential pipeline deployed on AWS.

Agent Decision Summary — Latest Run

Symbol	Market Signal	News Sentiment	Final Decision	Confidence

2. Data Collection

Data was collected for AAPL, MSFT, GOOGL (stocks) and BTC-USD, ETH-USD (cryptocurrencies) using the yfinance API. One year of daily OHLCV data was downloaded per symbol. Data is automatically uploaded to AWS S3 and stored in AWS RDS PostgreSQL via Lambda every weekday.

AWS Cloud Infrastructure

AWS Service	Configuration	Purpose
AWS S3	trading-raw-data-2024	Stores raw CSV files (stocks/ + crypto/)
AWS RDS	PostgreSQL, trading-db, us-east-1	Structured database for market data
AWS EC2	t2.micro, Amazon Linux 2023	Cloud computation and pipeline execution
AWS Lambda	Python 3.11, 256MB, 1min timeout	Automated daily data fetch
EventBridge	cron(30 13 ? * MON-FRI *)	Triggers Lambda weekdays 9:30AM EST

3. Exploratory Data Analysis

Technical indicators were engineered: MA20, MA50, EMA12, RSI (14-day), MACD with Signal and Histogram, and Bollinger Bands. Buy signals: RSI < 40. Sell signals: RSI > 60.

Technical Analysis Charts

Figure: AAPL — Price, MA, Bollinger Bands, RSI and MACD

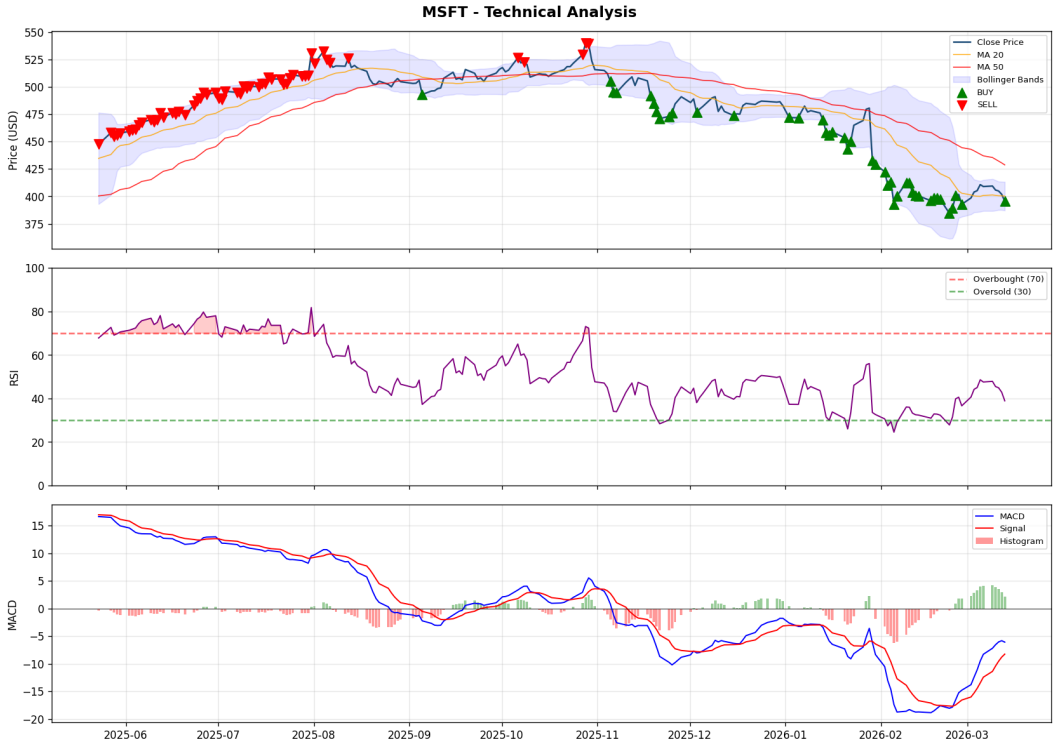

Figure: MSFT — Price, MA, Bollinger Bands, RSI and MACD

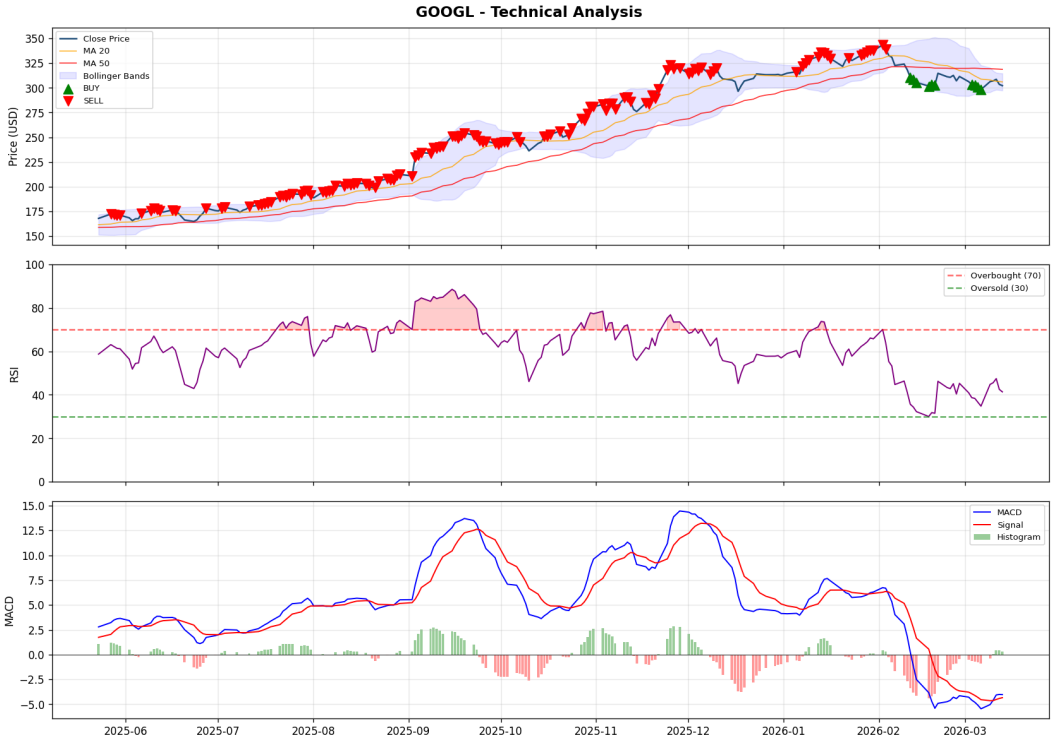

Figure: GOOGL — Price, MA, Bollinger Bands, RSI and MACD

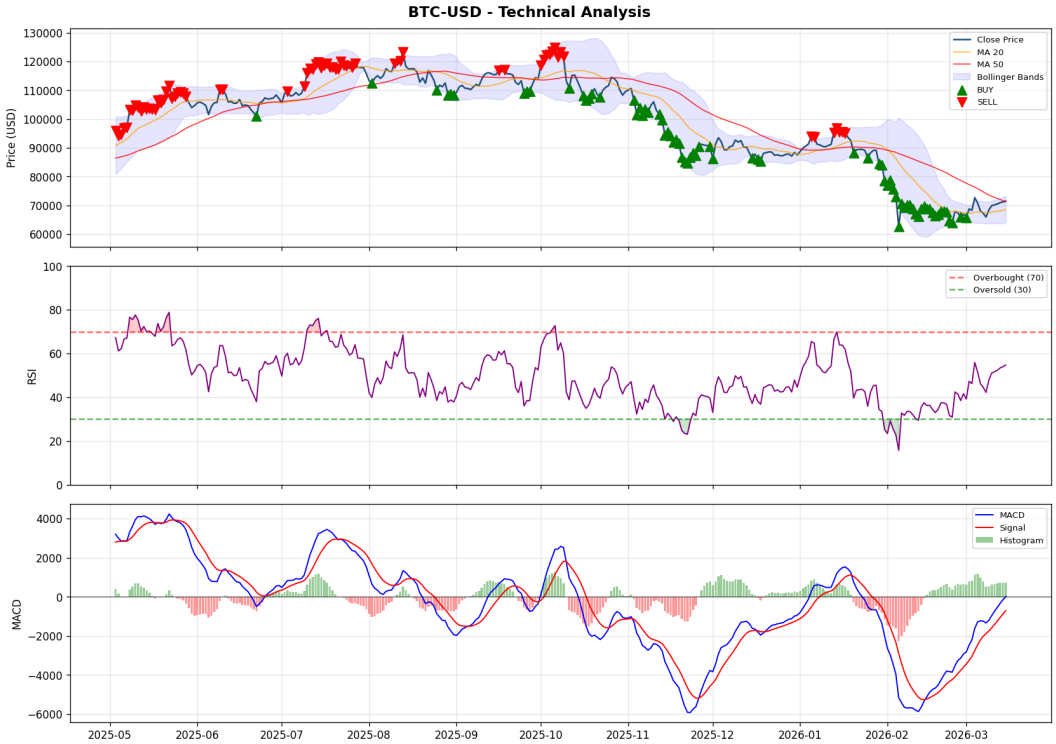

Figure: BTC-USD — Price, MA, Bollinger Bands, RSI and MACD

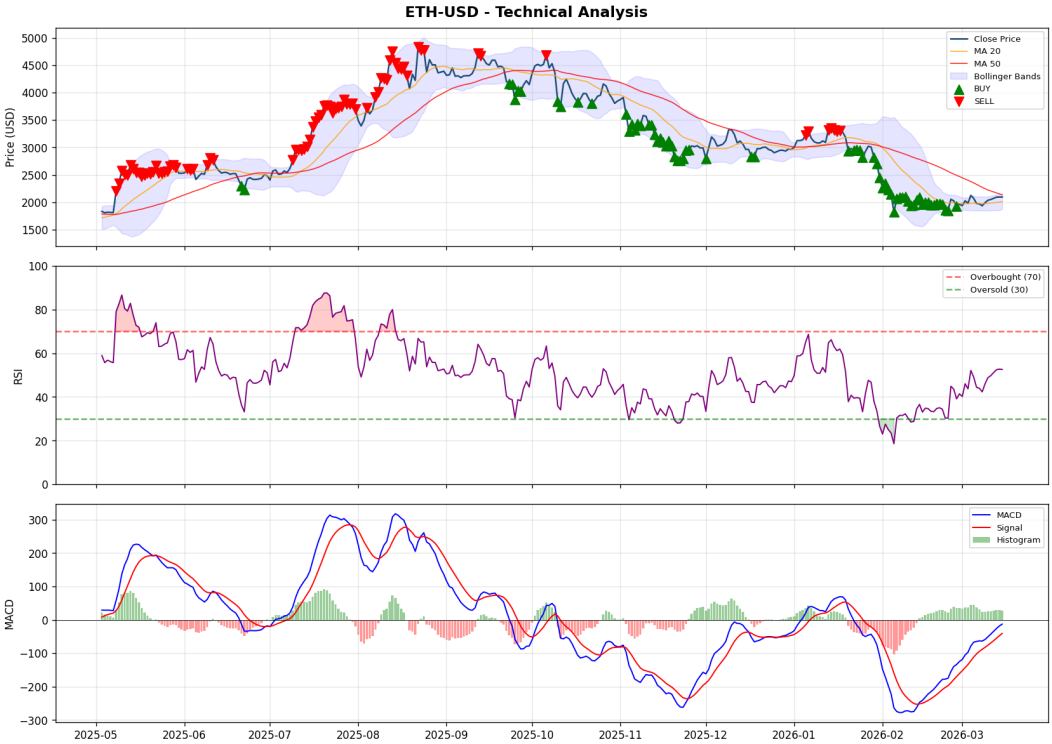

Figure: ETH-USD — Price, MA, Bollinger Bands, RSI and MACD

4. Correlation and Volatility Analysis

BTC and ETH show very high correlation (0.84), meaning they move together. Stocks and crypto show near-zero correlation, providing portfolio diversification benefits.

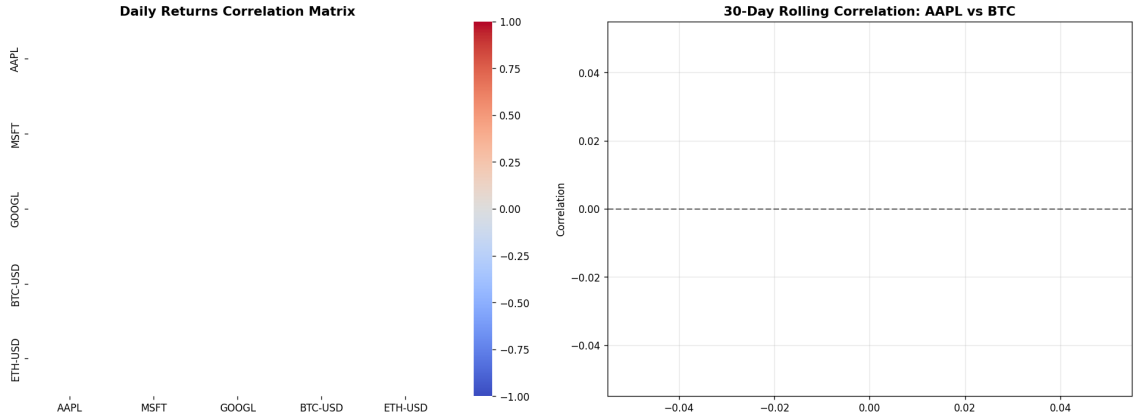

Figure: Correlation Heatmap and 30-Day Rolling Correlation (AAPL vs BTC)

Asset Pair	Correlation	Interpretation
AAPL vs MSFT	0.40	Moderate — both large-cap tech
AAPL vs GOOGL	0.50	Moderate — large-cap tech sector
BTC vs ETH	0.84	Very high — crypto moves together
AAPL vs BTC	0.00	No correlation — stocks vs crypto

Value at Risk — 95% Confidence

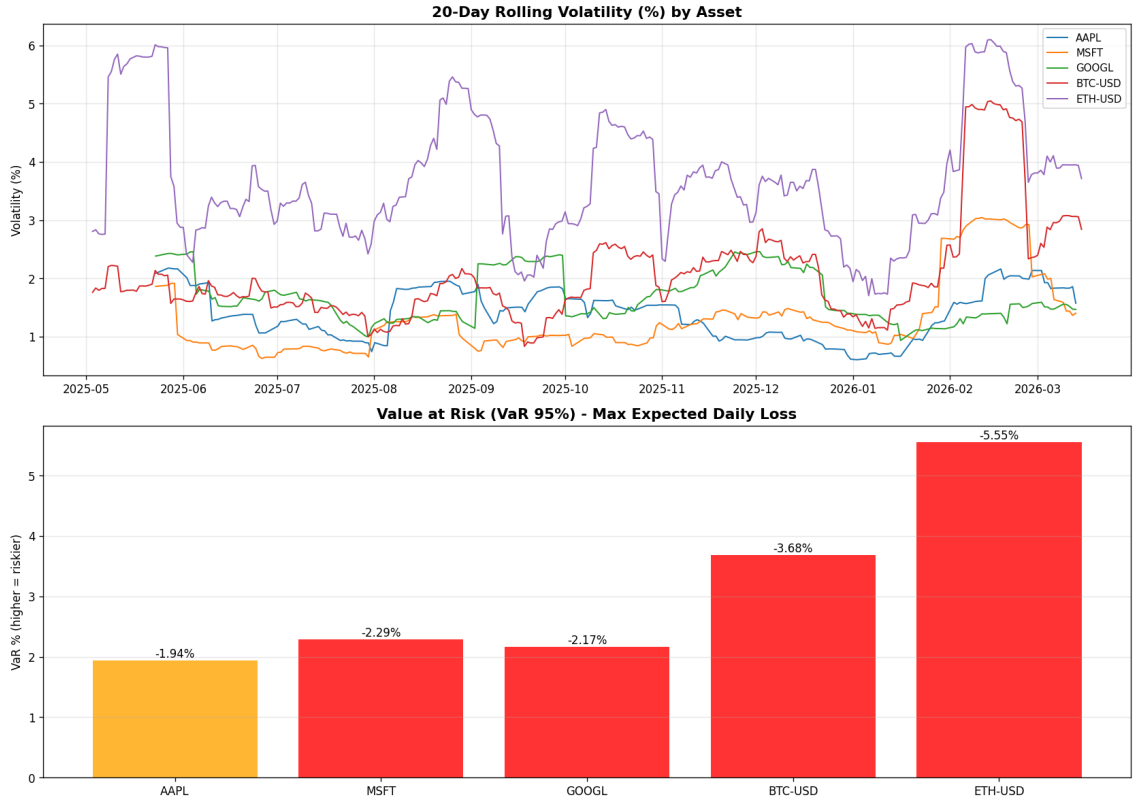

Figure: 20-Day Rolling Volatility and VaR (95%) per Asset

Asset	Daily VaR (95%)	Risk Level
AAPL	-1.94%	Lowest risk
GOOGL	-2.17%	Low risk
MSFT	-2.29%	Low risk
BTC-USD	-3.68%	Medium risk
ETH-USD	-5.55%	Highest risk

5. K-Means Asset Clustering

K-Means clustering (k=3) was applied using three standardised features: mean daily return, mean 20-day rolling volatility, and mean RSI.

Asset	Cluster	Group Interpretation
AAPL	0	Stable, low-volatility
MSFT	0	Stable, low-volatility
BTC-USD	0	Stable, low-volatility
GOOGL	2	Mid-volatility
ETH-USD	1	High-volatility crypto

6. Machine Learning Model

A Random Forest Classifier (100 estimators, max_depth=5) was trained per asset to predict Buy/Sell/Hold signals using 9 features. RSI was the dominant predictor with 46-63% importance across all assets.

Feature Importance Charts

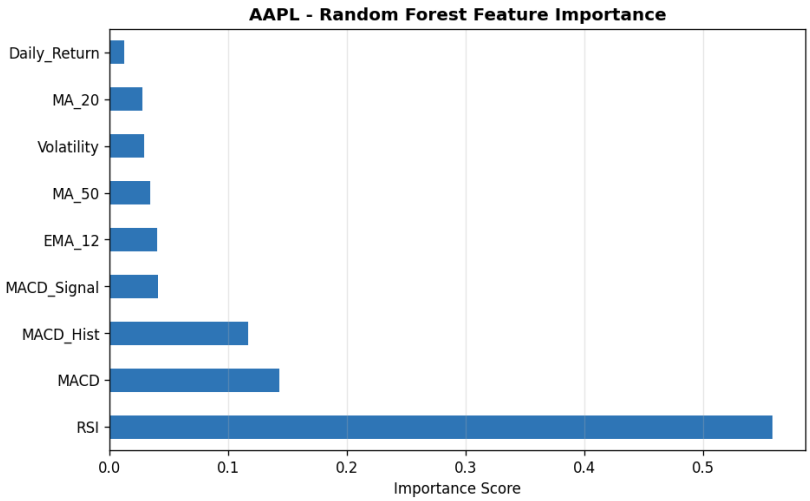

Figure: AAPL — Random Forest Feature Importance

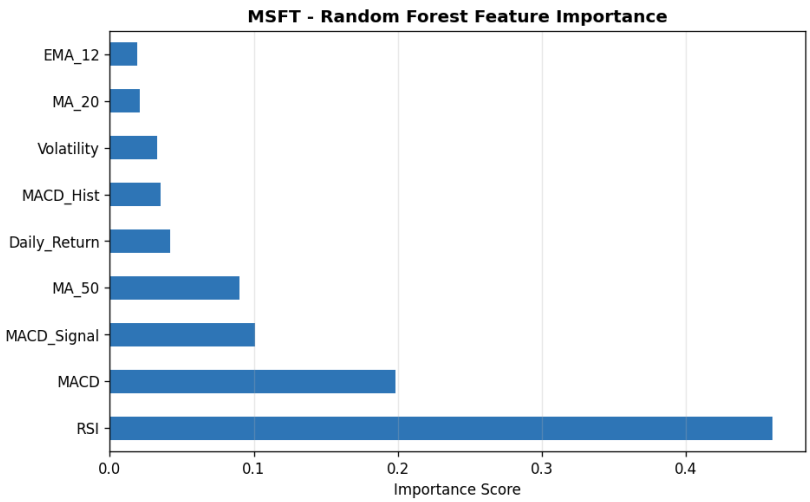

Figure: MSFT — Random Forest Feature Importance

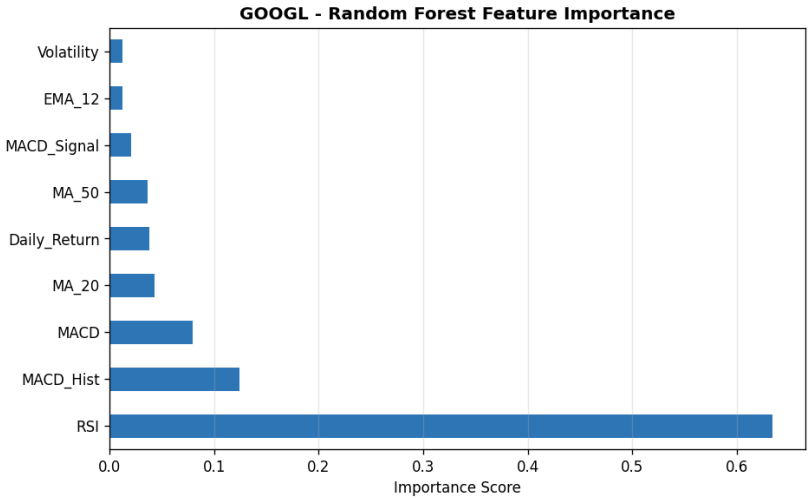

Figure: GOOGL — Random Forest Feature Importance

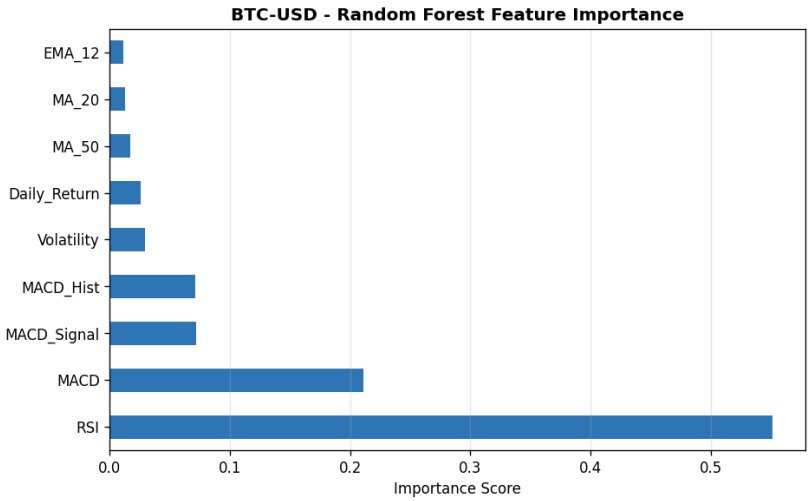

Figure: BTC-USD — Random Forest Feature Importance

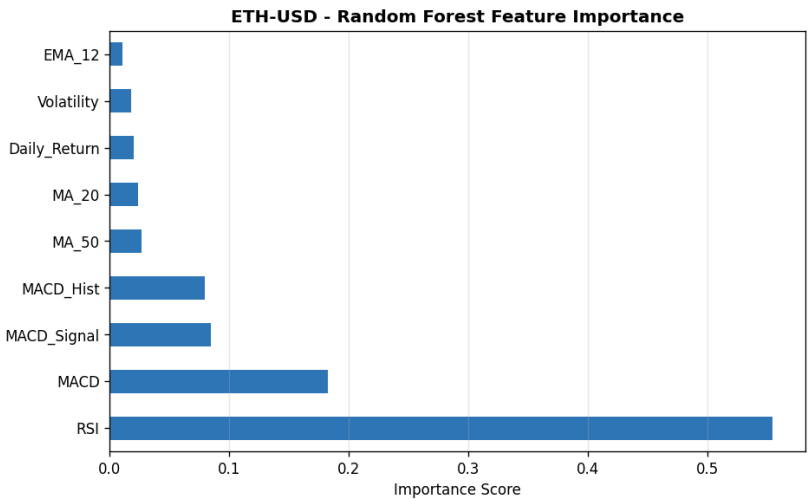

Figure: ETH-USD — Random Forest Feature Importance

7. Agentic AI Workflow

The system implements a four-agent sequential pipeline where each agent processes its input and passes results to the next agent.

Agent 1 — Market Analysis

File: market_analysis.py — Analyses RSI, MACD and Moving Average crossovers. RSI<40=BUY, RSI>60=SELL.

Agent 2 — Information Retrieval

File: info_retrieval.py — Fetches 15 headlines per symbol. Scores with VADER NLP (>0.05=POSITIVE, <-0.05=NEGATIVE).

Agent 3 — Decision Engine

File: decision_engine.py — Combines signal + sentiment. POSITIVE news boosts BUY confidence by +20%.

Agent 4 — Risk Manager

File: risk_manager.py — Applies risk controls: min 45% confidence, max 15% position, 5% stop-loss, max 5 trades/day.

8. Backtesting Results

Backtesting simulates historical trading using generated signals. Sharpe Ratio >1 indicates good risk-adjusted return. AAPL achieved the best Sharpe Ratio of 1.29.

Performance Summary — Latest Run

Symbol	Signal	Sentiment	Return	Sharpe	Drawdown	Win Rate	ML Acc.
AAPL	SELL	POSITIVE	+5.84%	1.29	-3.5%	100%	97.6%
MSFT	BUY	POSITIVE	-14.99%	-1.23	-18.3%	100%	100%
GOOGL	BUY	POSITIVE	-0.51%	-0.16	-2.6%	0%	75.6%
BTC-USD	BUY	NEUTRAL	+0.00%	0.0	0.0%	0%	100%
ETH-USD	BUY	NEUTRAL	-9.65%	-0.21	-26.8%	100%	100%

Backtest Charts

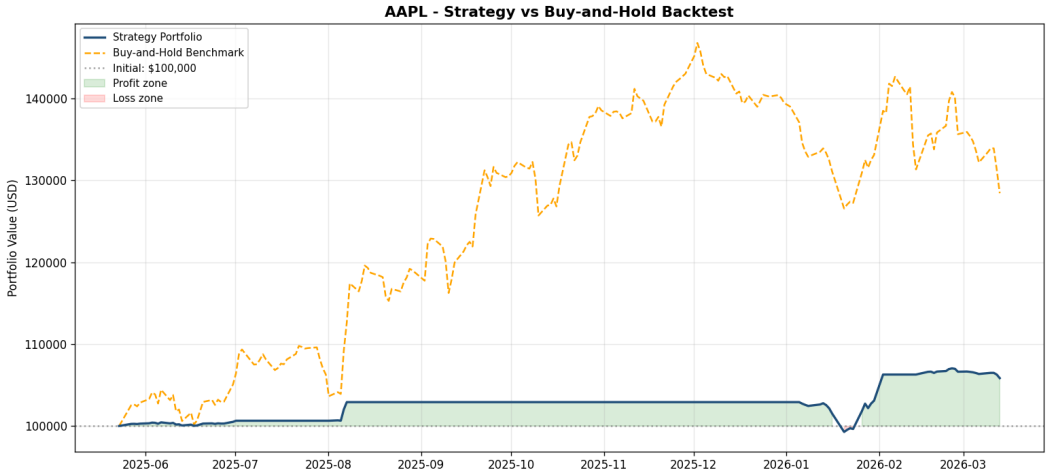

Figure: AAPL — Strategy Return vs Buy-and-Hold Benchmark

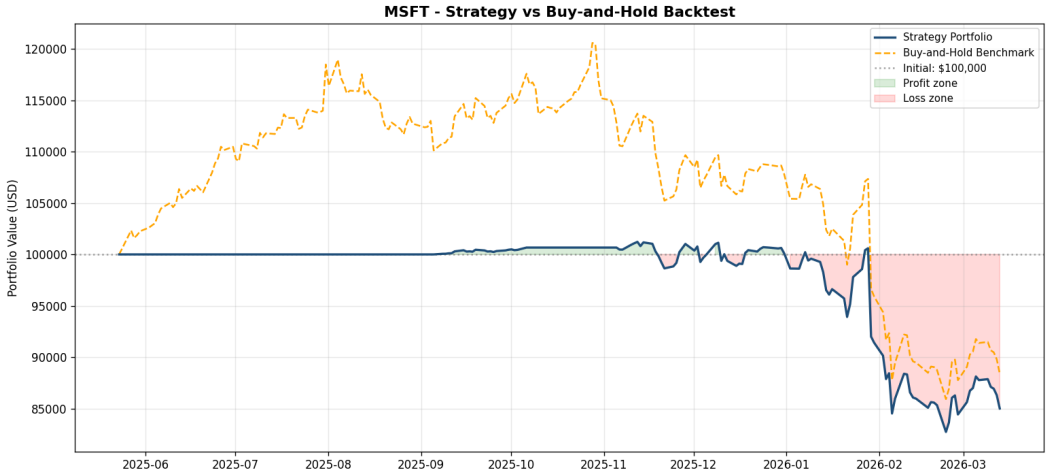

Figure: MSFT — Strategy Return vs Buy-and-Hold Benchmark

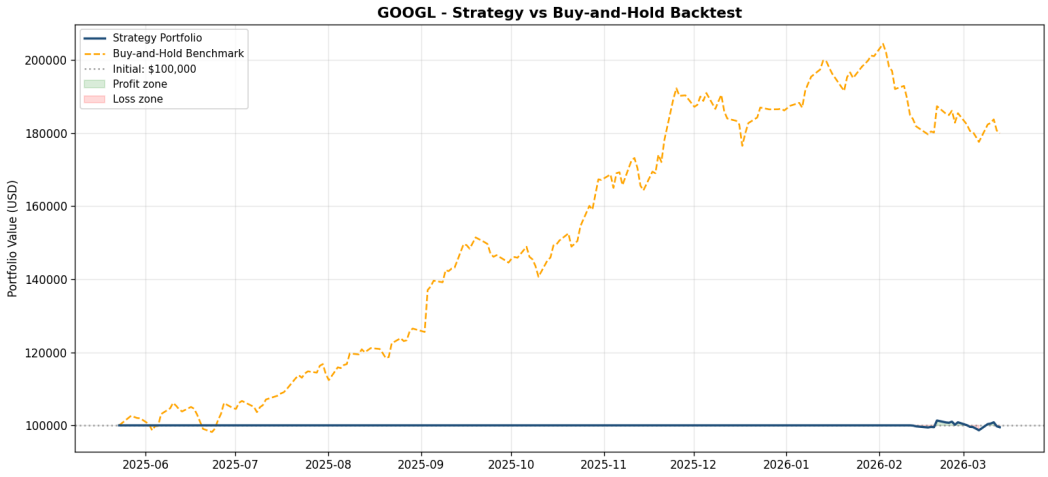

Figure: GOOGL — Strategy Return vs Buy-and-Hold Benchmark

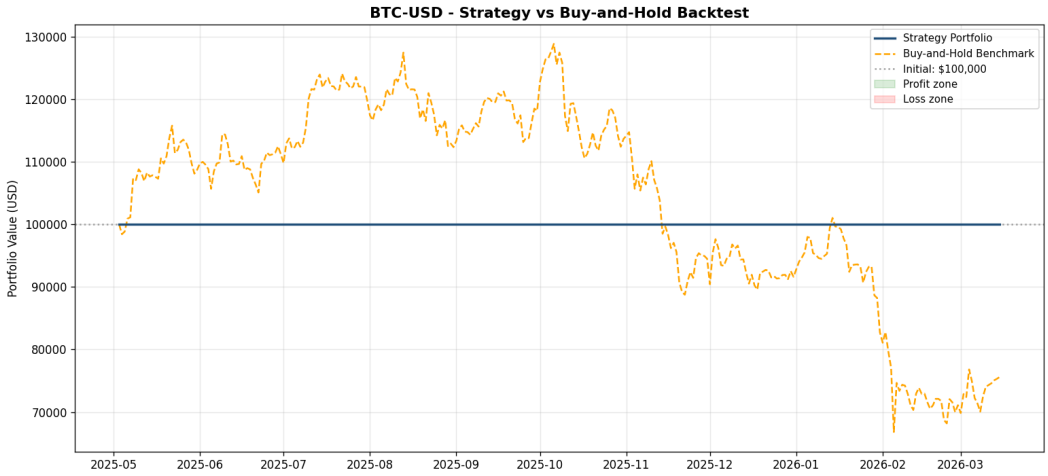

Figure: BTC-USD — Strategy Return vs Buy-and-Hold Benchmark

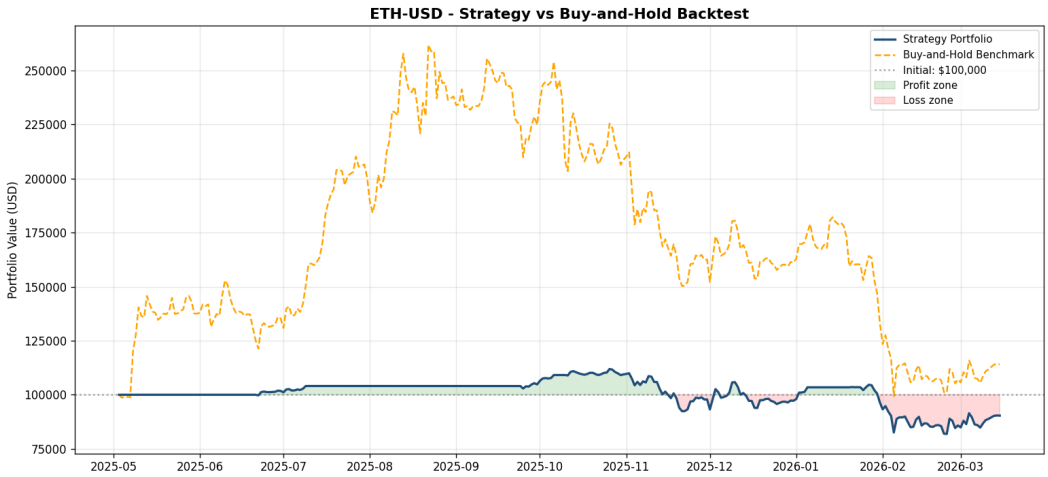

Figure: ETH-USD — Strategy Return vs Buy-and-Hold Benchmark

9. Business Impact

Automation: Lambda eliminates manual daily data collection — runs every weekday automatically.

Risk Management: Four-layer agent system prevents impulsive trades through confidence thresholds.

Diversification: BTC-ETH correlation 0.84 — portfolios should not over-allocate to both.

Risk Ranking: ETH-USD has 3x more daily risk than AAPL (VaR -5.55% vs -1.94%).

Scalability: Can be extended to more symbols or integrated with Alpaca/Interactive Brokers API.

10. Limitations

Backtesting Bias: RSI thresholds tuned on training data may overestimate live performance.

ML Overfitting: 100% accuracy on some assets reflects small test sets (41-64 samples).

NLP Model: VADER is general-purpose. FinBERT would give more accurate financial sentiment.

No Live Trading: System generates signals only — no real brokerage API integration yet.

Single Timeframe: Only daily data used. Intraday signals not currently supported.

11. Conclusion

This project successfully implemented a complete Agentic AI trading workflow covering all required components: Market Analysis, Information Retrieval, Decision Engine, and Risk Management. All infrastructure is deployed on AWS with automated daily execution via EventBridge. The ML model achieved 75.6-100% accuracy across assets. AAPL demonstrated the best risk-adjusted performance (Sharpe 1.29). Correlation analysis confirmed BTC-ETH move together (0.84) while stocks and crypto remain uncorrelated (0.00), providing clear portfolio diversification guidance.